

# Hritik Saynganthone

AKA Ricky, US Citizen

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## OBJECTIVE

To obtain a full-time position in software development, located either in-person/hybrid in and around Rochester, Syracuse, or Buffalo, NY. Alternatively, anywhere remote. Available starting immediately.

## EDUCATION

### Rochester Institute of Technology

M.S. in Computer Science, Minor in Mathematics

Member of the *Document and Pattern Recognition Lab*

Relevant Courses: *Machine Learning, Foundations of Computer Vision, Global Illumination, Introduction to Software Engineering, Concepts of Parallel and Distributed Systems, Computational Complexity*

Rochester, NY  
May 2026

## WORK EXPERIENCE

### Solü Technology Partners

Software Development Intern

Victor, NY  
Jan. '23 - Aug. '23

- › Worked with an established code base with other interns, utilizing Agile Scrum methodologies within the Atlassian suite of tools.
- › Utilized several JVM-based technologies for backend development of new web application features.
- › Interfaced with both MySQL and MongoDB databases for user and application data, respectively.
- › Created several new frontend features, applying MVC design patterns and component-based architectures using Angular.
- › Verified and produced several API endpoints using Postman, to be hit in various stages of the application lifecycle.
- › Unified and simplified software infrastructure by completely containerizing the entire development, testing, and production environments with Docker Compose and AWS ECR/ECS.
- › Introduced the use of automated tested via AWS CodePipeline for simultaneous backend, frontend, API, and system integration tests.
- › Technologies used: **Java, Kotlin, OOP, Maven, Gradle, Spring Boot, RESTful APIs, Mongock, MongoDB, MySQL, Angular, MVC, TypeScript, HTML, CSS, Docker, Docker Compose, Microservices Architecture, AWS CodePipeline, AWS ECR/ECS, AWS EC2, CI/CD, Python, Postman, Swagger, Selenium, Git, Trello, Agile Scrum, Confluence, BitBucket.**

Software Development Intern

May '24 - Dec. '24

- › Continued the development and upkeep of previously implemented systems and features.
- › Led the development of the Solü Application Service Tool, a Python-based command line tool that can be used by developers to quick instantiate different environments on any system they are using.
- › Initialized the development of a multi-tenant system, allowing a different backend system to be loaded with different data without needing a completely separate environment.
- › Technologies used: **Python, Bash, Spring Boot, SQL, Docker.**

Software Development Intern

May '25 - Aug. '25

- › Maintained the previously developed web application as the sole developer on the project.
- › Implemented an MVP Retrieval Augmented Generation system using LibreChat, such that company data could be stored in a vector database and an LLM could be queried for information therein.
- › Technologies used: **Python, Azure AI Foundry, Azure CosmosDB, LibreChat, ChatGPT.**

## WORK EXPERIENCE (CONT.)

Kate Gleason College of Engineering

Research Assistant

Rochester, NY  
Jan. '22 - May '22

- › Designed and developed the Hybrid Learning Optimization Solver Architecture, used to wire several mathematical optimizers together for finding approximations to NP-hard problems.
- › Instituted a client-server oriented API to interface with multiple solvers, including both traditional algorithms and machine learning algorithms.
- › Technologies used: **C#, Java, Python, .NET, Apache Kafka, FastAPI, Git.**

## ADDITIONAL PROJECTS

Master's Thesis

Jan. '24 - May '26

- › Conducted novel research within the machine learning field of Optical Music Recognition, submitted for both Master's Thesis approval and the International Journal on Document Analysis and Recognition.
- › Explored high-entropy alternatives to the Connectionist Temporal Classification (CTC) loss function, utilizing convolutional recurrent neural networks (CRNNs).
- › Technologies used: **Python, Conda, Virtualenv, PyTorch, TensorFlow, CUDA, Music21, NumPy, OpenCV, Matplotlib, Git, Gitlab, Github, Linux, LaTeX, Slurm.**

Personal Website Built in Django

June '26

- › Used a lack of personal website as a reason to learn Django and get better at raw HTML/CSS.
- › Utilized Docker Compose and Cloudflare tunnels to create completely self-host the website on my own personal server.
- › Technologies used: **Python, Django, PostgreSQL, Docker Compose, Docker, Gunicorn, Virtualenv, Makefile, Nginx, Cloudflare Tunnels, HTML, CSS, Git.**

Airspace Sectorization via Voronoi Diagrams

Aug. '25 - Dec. '25

- › Developed an algorithm to sectorize a given airspace using a voronoi diagram based on restrictions simulated via a genetic algorithm.
- › Technologies used: **Python, Genetic Algorithms, Voronoi Diagrams, OpenSky, FastAPI, NumPy, Git, GitHub.**

From Scratch Raytracing Engine

Jan. '25 - May. '25

- › Developed multithreaded raytracing engine that can render simple objects within 3D space.
- › Technologies used: **C#, Thread Pools, Git, GitHub.**

Full Stack E-Store Web Application

Jan. '22 - May '22

- › Utilized Java Spring Boot & Angular in order to uphold key object-oriented design principles.
- › Employed the full Agile development cycle with a team of software developers from start to finish.
- › Technologies used: **Java, Spring Boot, RESTful APIs, Angular, Agile Scrum, Trello, SonarQube, Git, GitHub.**

Depth First Search Sudoku Solver

July '21

- › Independently created a Java interface and class file to solve DFS-compliant puzzles.
- › Formulated and executed a way to take in 9x9 sudoku puzzles and output a solution as a command line application.
- › Technologies used: **Java, Git, GitHub.**

## ADDITIONAL HOBBIES & INTERESTS

- › Member (2y), Wind Section Lead (1y), and Conductor (3y) of the RIT Game Symphony Orchestra.
- › Active musician in both classical and jazz performance.
- › Avid collector of CDs and soundtracks.
- › Home audio enthusiast!
- › Huge coffee person with a nice hand grinder and multiple different brewers.
- › Avid gamer — ask me what I'm currently playing!